Intelligent OpenShift Systems Management and Control

As systems need to become more reliable while increasing in complexity, a self-organizing machine with the ability to be monitored intersecting its decision-making process is a strong solution. The problem is that, eventually, OpenShift systems will become too complex for human teams to manage them properly.

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It is a widely-accepted view that control systems like containerization grouped into clusters given their orient in computing environments is the *de facto* method. This transition can be clearly illustrated in organizations evolving from CD v1 to a more loosely-coupled CD v2.

While the move to applications in container "farms" and the addition of bazel toolsets for distroless imaging mitigates complexity in build and deployment of developer apps, the next-step is to attending to the monitoring.

Monitoring, tuning, and optimization are the most requisite processes to contemplate as since we have decided how to best develop and deploy our customer's applications, given their necessary configuration, leaves us with how to judge performance, tune the cluster to tweak moving toward a state of systematic equilibrium, where the platform is essentially self-guiding.

This white paper describes a technology that has an OpenShift system moving toward systematic equilibrium by leveraging AI written through the lens of cybernetics.

Introduction

This organization is forward-looking to absorbing its future impacts by smartly investing in the devlopment of technologies learning toward an OpenSource posture. We have achieved great things by the implementation of distroless as a key component of CD2.0; with the increasing complexity of system evolution, given that developers have the power to have their programs express meaning much more effectively than with CD1.0, it is inevitable to assume this complexity should be addressed sooner than later.

In the current configurations of OpenShift, there are many opportunities to collect performance data and extrapolate meaning of how the system behaves over time in order to reflect the right container resources (cpu, memory, storage) mitigated by operations to "find" the right balance. In common practice, discovery of this changing cumulative metric, a differential equation, is done by hand and not by a learning algorithm. As systems grow in complexity, monitoring of this complex state becomes more critical as the guesswork of the past no longer applies to the model.

The offering, informally called an *aeon*, is a disembodied form of a guise. A guise is a template personality that is tasked to perform certain complex analyltical work on cluster performance data first looking for trends, but then upon compartmentalization by our own patented techniques, the aeon adapts and can self-regulate the clusters it is tasked with while aiding in disaster mitgation in times of emergency. Over time, by remembering and incorporating new data into its computations, it can also inform humans of what it is doing or if it should request a decision confirmation.

Maintenance of this is accomplished by its architecture: It is an application provided as a container that can be deployed into the cluster it is to analyze, optimize, or dry-decision-run metrical data. Configuration is done as a part of deployment; how much autonomous control is staged so that humans can see and become more confortable with how the aeon works and responds. A dialogue-based terminal is available in the container where engineers can ask it humanlike questions: "How has server C01-03D peformed since the last analysis?" and "Can we adjust C02-02P to prefer performance over data-checking?".

It is inevitable that organizatinos using OpenShift will hit by this giant technological stumble. People who believe that people can solve this problem are completelyl correct: those who believe that humans can solve it technically are wrong and putting their systems at risk.

What means intelligent OpenShift?

Equilibrium the figure.

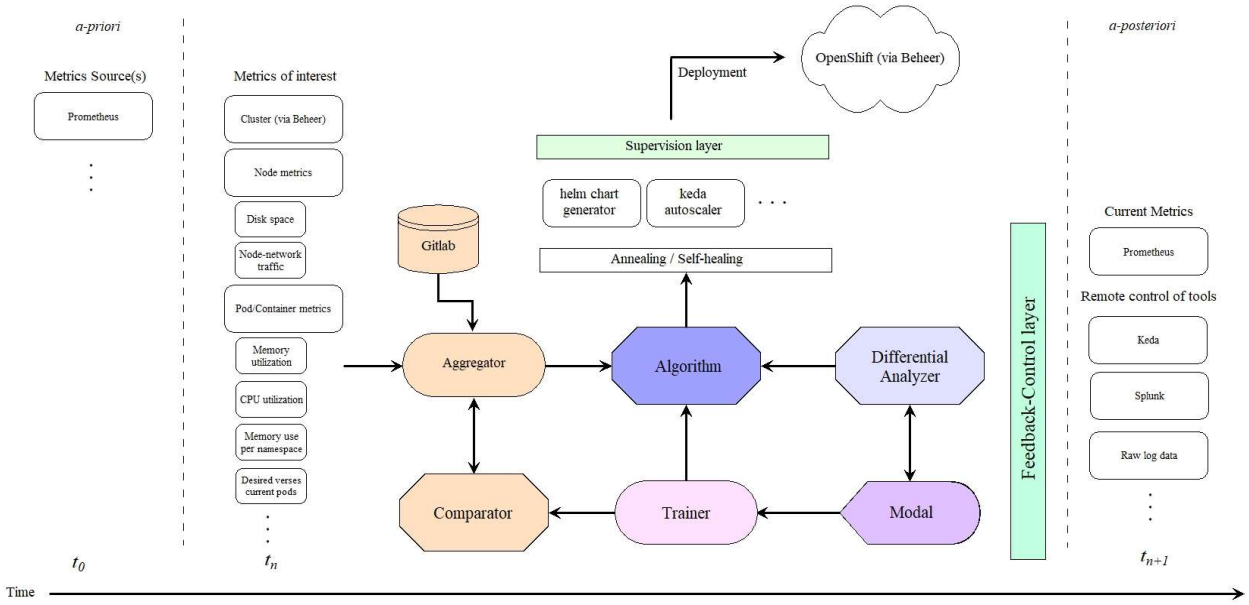

Fig. 1: A model of autonomous control.

Words.

The intellectual foundations of this technology lies in the domain of cybernetics, outside of comptuer science. This is important, as people trying to design by computer science will miss many critical points by having access to research in behaviourial problems outside the academic scope of understood scope and application of computable problems. Our particular contribution is the introduction of a feedback-weighted agent software with an interactive interface consequentially trained and reinforced on live metrics from the servers it is set to pay attention where the workflow patterns create a model as an artificial structure. The model is defined in such a way that its approach to reasoning about its environment *a priori* will yield the method of analysis for an interpretation *a posteriori* of meaning resembling a unique execution of pathways resulting in its reasoning facilities, forming an *understanding* how the clusters behave over their lifetime and can therefore optimize them toward operational equilibrium.

Here at Cartheur, we are very excited about this proposal. It has been a challenging area of research that has taught us not only about humanity but the concept of life in a very general sense. A lot of research, thought, expression, and love have gone into developing emotional toys and we hope you enjoy interacting and living with them as we had in designing and building them.